

David Manuel

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of enterprise experience architecting cloud-native microservices, event-driven streaming pipelines, and API integrations. Expertise in Akeneo PIM SaaS/Enterprise, high-volume catalog/SKU hierarchy modeling, distributed streaming with Apache Kafka and RabbitMQ, and multi-cloud infrastructure (AWS, GCP, Azure). Proven track record optimizing high-throughput database queries across PostgreSQL, MySQL, MongoDB, and Redis, delivering resilient, low-latency microservices under 24/7 high-availability standards.

CORE COMPETENCIES

PIM & Catalog Systems: Akeneo PIM (SaaS/Enterprise), Product Hierarchy Modeling, Variant Structures, Reference Entities, Category Mappings, Delta-Calculation Engines

Backend Frameworks & Languages: Node.js, TypeScript, Java (Spring Boot, Spring Cloud), Python (FastAPI), Go, C# / .NET Core

Streaming & Messaging: Apache Kafka (Topic Architecture, Consumer Groups, Partition Strategies, Schema Registry), RabbitMQ, IBM MQ

APIs & Security: RESTful, SOAP, GraphQL, gRPC, OAuth2, JWT, Enterprise SSO, RBAC, API Gateway Security Frameworks

Databases & Caching: PostgreSQL, MySQL, Oracle, CloudSQL, MongoDB, Redis, Elasticsearch

Cloud & DevOps: AWS (EKS, ECS, Lambda), GCP (GKE), Azure, Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Gradle, WebLogic, Tomcat

Observability & Tools: Grafana, Splunk, Prometheus, Datadog

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Dec 2022 – Present | Rochester, NY

Netsmartz

- Architected and evolved resilient, cloud-native backend microservices and stream-processing pipelines using Apache Kafka (schema registry, partition strategies), Node.js/TypeScript, and Java Spring Boot.
- Integrated custom catalog workflows, delta-calculation engines, and RESTful / SOAP APIs into distributed enterprise architectures, reducing median API latency from 850ms to 380ms (55% improvement).
- Managed and optimized multi-database environments across PostgreSQL, MySQL, MongoDB, and Redis for high-volume transactions, vector workloads, and memory-cached query acceleration.
- Deployed containerized workloads across AWS (EKS) and GCP (GKE) using Docker, Kubernetes, Terraform, and ArgoCD, ensuring 99.99% system availability.
- Established Grafana and Splunk observability dashboards and alerts for proactive system monitoring, reducing incident resolution times by 40%.

Senior Full Stack Software Engineer

Feb 2020 – Nov 2022 | Chicago, IL

Tempus AI

- Engineered secure, high-throughput microservices using Node.js, TypeScript, Python (FastAPI), and Java, backing enterprise catalog models and data processing engines.
- Implemented event-driven message queuing with RabbitMQ and Apache Kafka to handle real-time data syncs, schema migrations, and delta loads across distributed platforms.
- Optimized database indexing and query execution plans on PostgreSQL and MySQL, improving p95 query latency by 24% across high-volume datasets.
- Enforced enterprise security protocols across all API gateways using OAuth2, JWT, Enterprise SSO, and Role-Based Access Control (RBAC).
- Containerized services and automated multi-environment deployment pipelines using Docker, Kubernetes, and GitHub Actions.

Software Engineer

Dec 2015 – Jan 2020

Talmate

- Developed scalable backend services and REST APIs using Java (Spring Boot) and Node.js to integrate complex enterprise workflows and third-party systems.
- Built memory-caching solutions using Redis and optimized relational queries on MySQL and Oracle, reducing workflow execution times from 12 minutes to 5.5 minutes (54% speedup).
- Designed automated CI/CD build scripts using Gradle, GitHub Actions, and Azure DevOps across containerized environments.
- Participated in architectural reviews and mentored team members on clean code standards and Agile methodologies.

EDUCATION

Bachelor's Degree in Computer Science

Aug 2011 – May 2015 | Houston, TX

University of Houston